

Classification of MET Art by Time Period from Image Data

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Isaiah Lucero, Daria Slowinska
(Project 3 Leader), Carly Farrey
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Outline:

- Motivation & Hypothesis
- Research Question, Modeling Approach, and Goal
- Data Explanation and Acquisition
- Analysis Plan and Justification
- Tricky Analysis Decision
- Bias and Uncertainty Validation
- Results and Conclusions
- Next Steps
- References and Resources

Motivation:

- Classifying artwork by period is usually manual and subjective
- Large museum datasets make manual labeling inefficient
- Machine learning can learn visual patterns associated with time periods

Applications:

- Museum cataloging and archival automation
- Detecting mislabeled or unknown-period artwork



Hypothesis: A CNN can accurately classify paintings by historical time period.



Research Question:

How accurately can a machine learning model classify paintings into their historical time period based on image features?

Modeling Approach:

- Extract images from links, assign them unique names, and sort them into folders based on class.
- Clean the data and normalize images
- Split data into train and test sets and apply a CNN
- Evaluate model using model accuracy and loss

Goal: Classify paintings from the Metropolitan Museum of Art by its time period and analyze the precision of the model.

Data Acquisition and Explanation

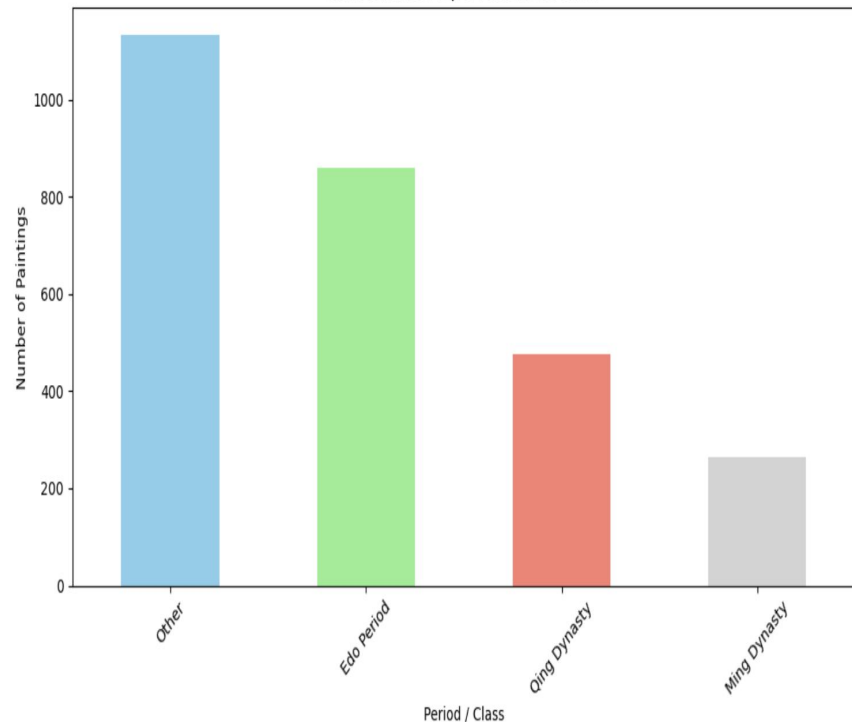
Column	Description
Title	Name of the painting
Artist Display Name	The name of the artist who created the painting
Period	What historical time period the painting is from
Object Date	A more specific time when the painting is from
Classification	The type of art
Link Resource	Link to the art piece on the Met website

Summary of Dataset:

- 6 columns:
 - Title
 - Artist Display Name
 - Period
 - Object Date
 - Classification
 - Link Resource
- 2,731 rows of data
- Origin: Metropolitan Museum of Art Website

Analysis Plan

Class Balance: Top 3 Periods vs. Other



Steps:

- 1) Collect images and period labels
- 2) Clean data and remove duplicate links
- 3) Resize and normalize images
- 4) Split into training and test sets
- 5) Train CNN to classify paintings by period
- 6) Evaluate using model accuracy and loss

Challenges Faced:

- Class imbalances
- Missing image links
- Inconsistent image size and quality
- Similar painting styles

Tricky Decision Analysis

Decision:

- Combining the Qing and Ming periods

Why it was Tricky

- We ran into many paintings classified as both “Qing or Ming” by the MET
- We also saw during the EDA that the classes were unbalanced

Impact of the decision

- We combined these classes to make the classes more balanced to avoid one class being too large, and to avoid classifying paintings incorrectly

Bias and Uncertainty Validation

Biases Considered

- Class imbalance between periods
- Poor image quality and resolution
- Overrepresentation of certain artists or styles
- “Other” class containing mixed periods

Uncertainty in Results

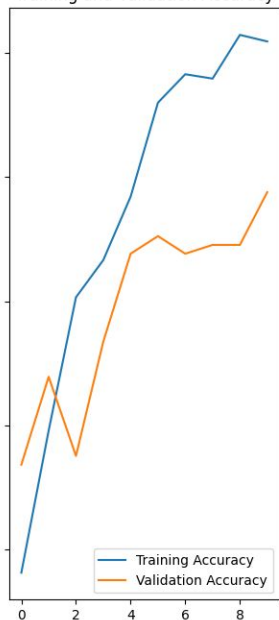
- Similar painting styles across periods
- Limited number of images for some classes
- Model sensitivity to preprocessing choices

How Bias Was Addressed

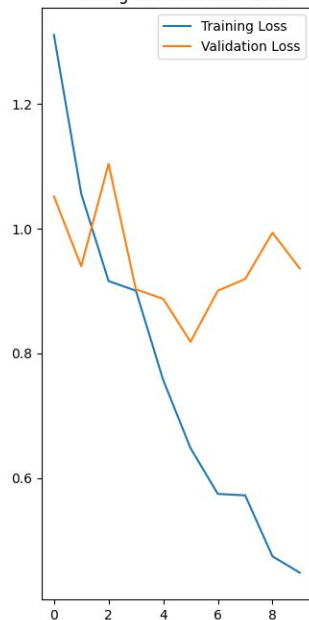
- Selected periods with largest sample sizes
- Checked class balance before training
- Used train/test split to reduce overfitting

Results and conclusions

Training and Validation Accuracy



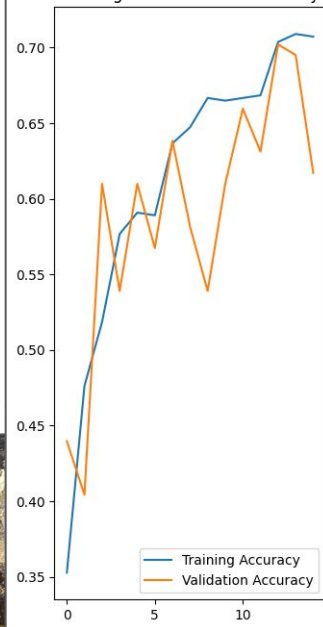
Training and Validation Loss



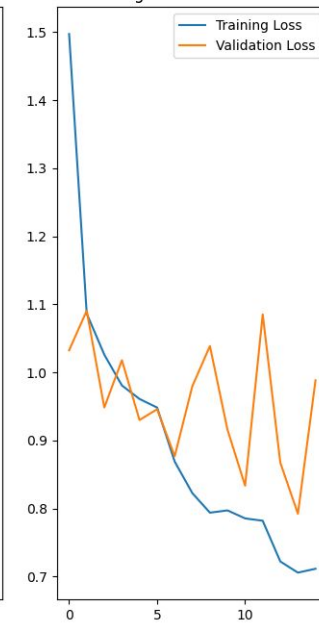
Default Model

Data Augmentation Model

Training and Validation Accuracy



Training and Validation Loss



Conclusion:

- Accuracy generally improved with more epochs
- Validation accuracy was much worse than training accuracy → overfitting
- Data augmentation slightly reduced validation loss
- The trained model was able to reach validation accuracy of **~70%**

Next Steps

New Lines of Exploration:



- Classify paintings by artist
- Expand dataset with more types of art
- Analyze which visual features drive predictions

Improvements:



- Increase dataset size for more training and validation data
- Have better class balance
- Use a pretrained model

New Questions:



- Which features are most important for classification?
- Does image resolution affect performance?
- Can the model generalize to other museums?

References

[1] McKillop, T. (2022, July 5). *What is image classification?* The Data School.

<https://www.thedataschool.co.uk/tyler-mckillop/what-3/>

[2] Strezoski, G., & Worring, M. (2017). OmniArt: Multi-task deep learning for artistic data analysis. arXiv.

<https://doi.org/10.48550/arXiv.1708.00684>

[3] Tensorflow, "Image Classification," *Tensorflow*. [Online]. Available:

<https://www.tensorflow.org/tutorials/images/classification>

Github: <https://github.com/dsizzle67/Project3-DS4002>

Questions?

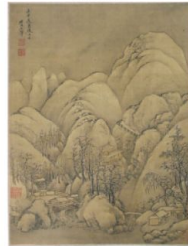
Qing Dynasty
ID: 35969



Ming Dynasty
ID: 35970



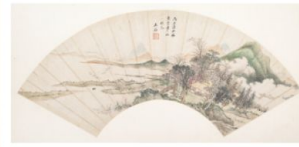
Qing Dynasty
ID: 35971



Qing Dynasty
ID: 35972



Qing Dynasty
ID: 35973



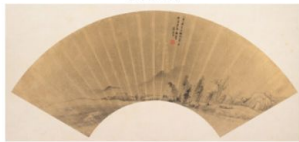
Qing Dynasty
ID: 35974



Qing Dynasty
ID: 35975



Other
ID: 35977



Other
ID: 35979



Other
ID: 35981



Ming Dynasty
ID: 35984



Other
ID: 35986



Ming Dynasty
ID: 35988



Ming Dynasty
ID: 35989



Ming Dynasty
ID: 35990

